

Data Analyst Capstone: Technology Trends

Allen Stalcup • January 14, 2026



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OUTLINE



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- • Slides 8–9 — Database Trends
- • Slide 10 — Job Postings Snapshot
- • Slides 12–14 — Dashboards (Current, Future, Demographics)
- • Slide 15 — Dashboard Insights
- • Slide 16 — Overall Findings & Implications
- • Slide 17 — Conclusion
- • Slide 18 — Appendix
- Slides 19–30 — Extended appendix (notebook EDA + supporting visuals)
- Slides 31–33 — Next steps (recommendations, roadmap, closing)

EXECUTIVE SUMMARY



- **CORE STACK** (highest ROI) : JavaScript/TypeScript + SQL + Python remain Top-5 in both current usage and next-year demand.
- **MODERNIZATION** (edge signals) : TypeScript demand > usage; Go/Rust enter the desired Top-10; Redis rises for real-time systems.
- 🧠 • **WHAT TO DO:** Build SQL + PostgreSQL + Python + BI, add cloud basics (AWS), then specialize (Product: JS/React; Data: modeling; Backend: Go/Rust).

INTRODUCTION



- **Purpose**
 - Compare current vs next-year technology adoption (languages + databases) and translate results into actionable skill recommendations.
- **Target audience**
 - Hiring managers, learners/career switchers, and educators planning upskilling or curriculum.
- **Key questions**
 - What tools dominate today?
 - Which tools are most desired next year?
 - What skill roadmap is the smartest investment?

METHODOLOGY

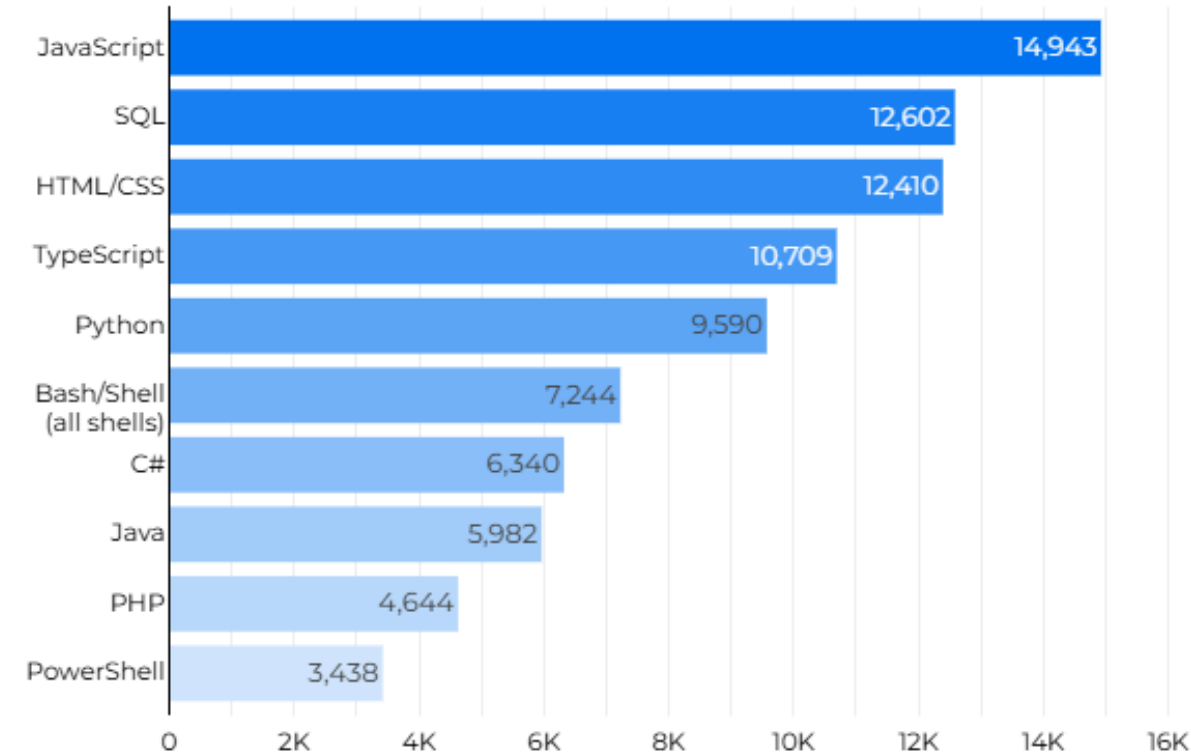


-  **Data sources**
 - Stack Overflow 2024 survey (survey_data_updated.csv) for technology + demographics.
 - Job market artifacts: Job API extract (job-postings.xlsx) + scraped salary by language (popular-languages.csv).
-  **Wrangling**
 - Multi-select fields split and exploded to long format; standardized labels; removed blanks/nulls.
 - Computed respondent counts (and % where applicable), selected Top-10, sorted descending for comparability.
-  **Tools**
 - *Looker Studio/Cognos for dashboards; optional Python/pandas for reproducible wrangling + checks.*
-  **Quality checks / limitations**
 - Counts reflect mentions (not mutually exclusive); survey sample bias noted in Appendix.
- **Limitation:** *Results reflect self-reported survey responses and may not represent all labor markets equally.*

PROGRAMMING LANGUAGE TRENDS

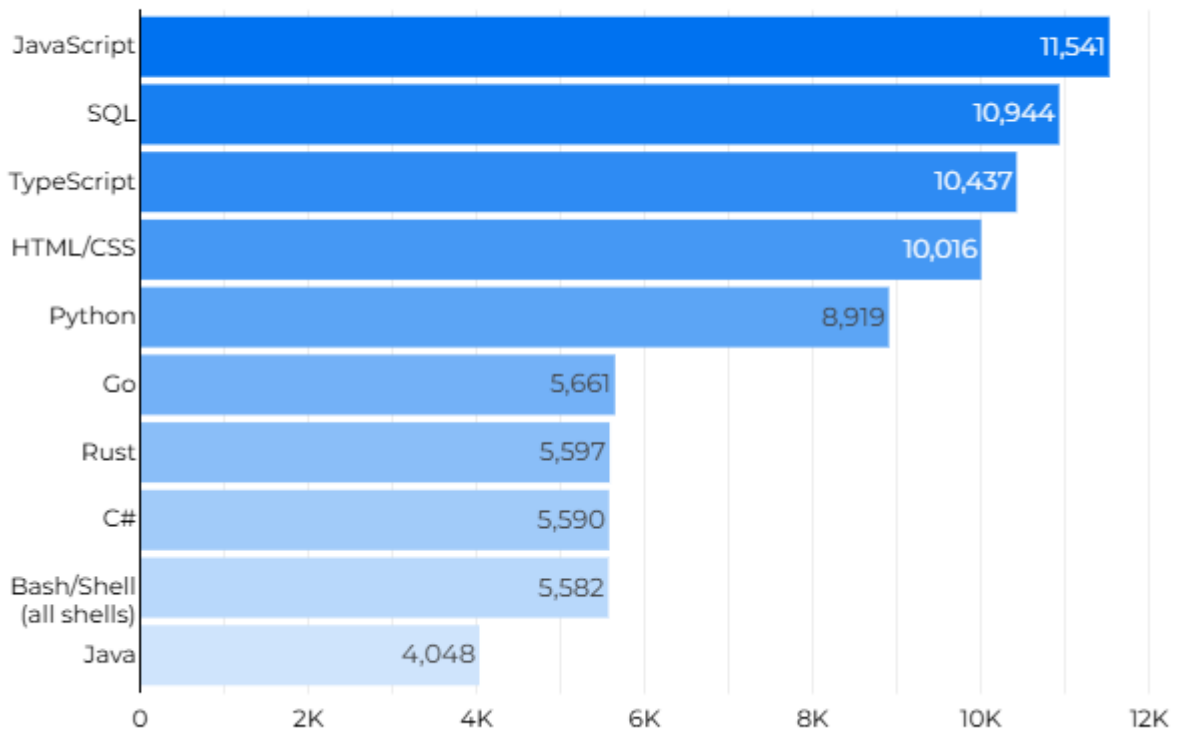
Current Year

Top 10 Languages Worked With



Next Year

Top 10 Languages Desired Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

-  Stable core: JavaScript, SQL, HTML/CSS, TypeScript, Python lead both current + desired.
-  Modernization: TypeScript demand > usage → teams standardizing typed JS for maintainability.
-  Growth bets: Go and Rust enter “Desired Next Year” → cloud-native + performance interest.
- Impact
- **Data/Bi:** SQL + Python is the fastest employability lever.
- **Product analytics:** add JS/TS to work closer to web/app teams.
- **Differentiation:** learn one growth bet after the core (Go/Rust).

Recommendations

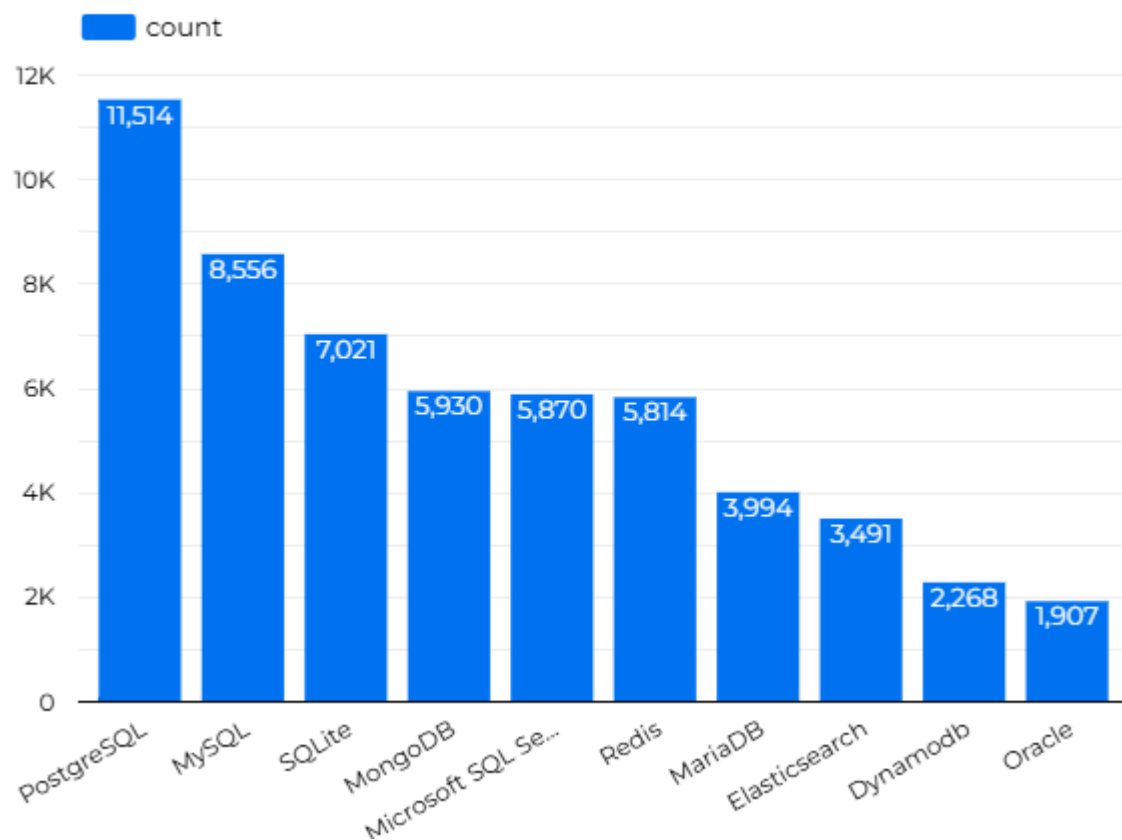
-  Learners: ship 1–2 projects proving SQL + Python + dashboarding (reproducible + documented).
-  Employers: invest in TS + managed data platforms and expect baseline cloud literacy (AWS/Azure/GCP).
-  Educators: emphasize end-to-end workflows (data → analysis → dashboard → decision memo).



DATABASE TRENDS

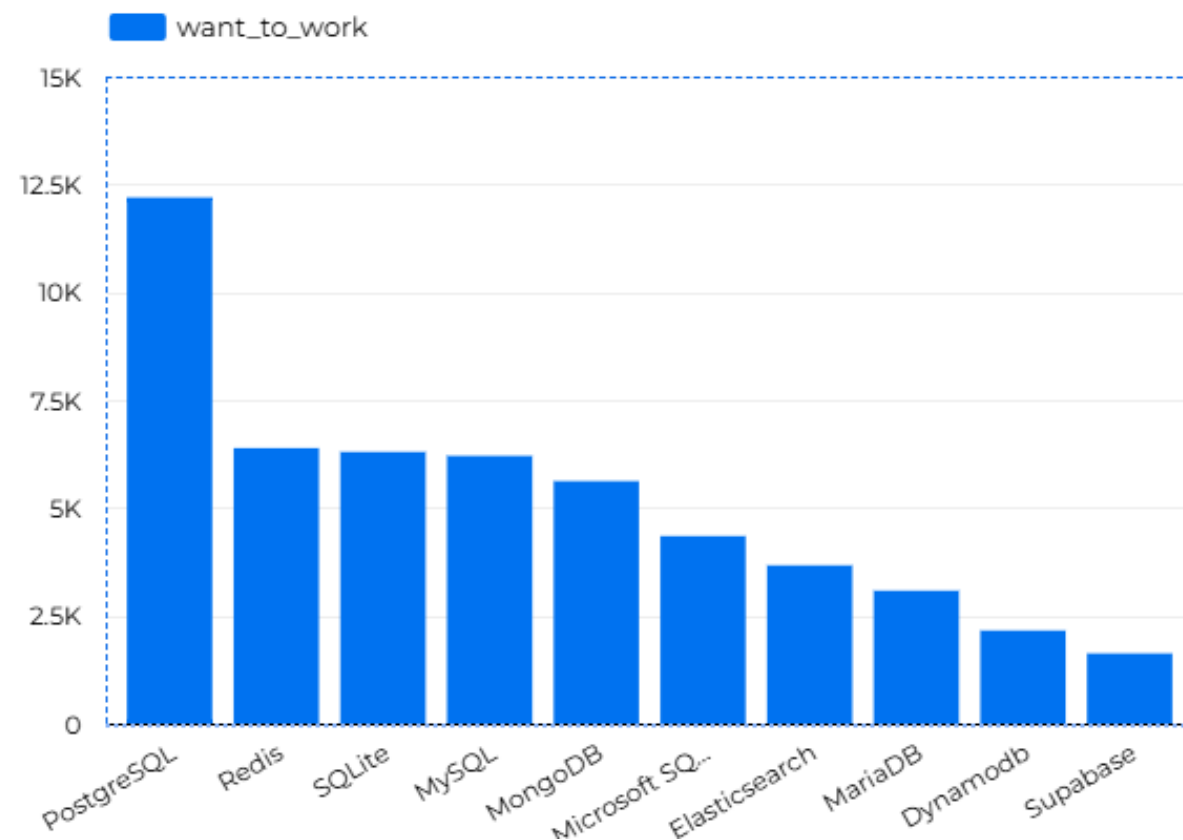
Current Year

Top Databases Worked With



Next Year

Top 10 Databases Desired Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

-  • PostgreSQL leads both current usage and future desire → default modern SQL database.
-  • MySQL + SQLite remain durable (web + embedded/local workflows).
-  • Redis rises sharply in desire → more real-time apps + caching patterns.
- • Mixed Top-10 (relational + NoSQL) shows fit-for-purpose persistence.

Recommendations

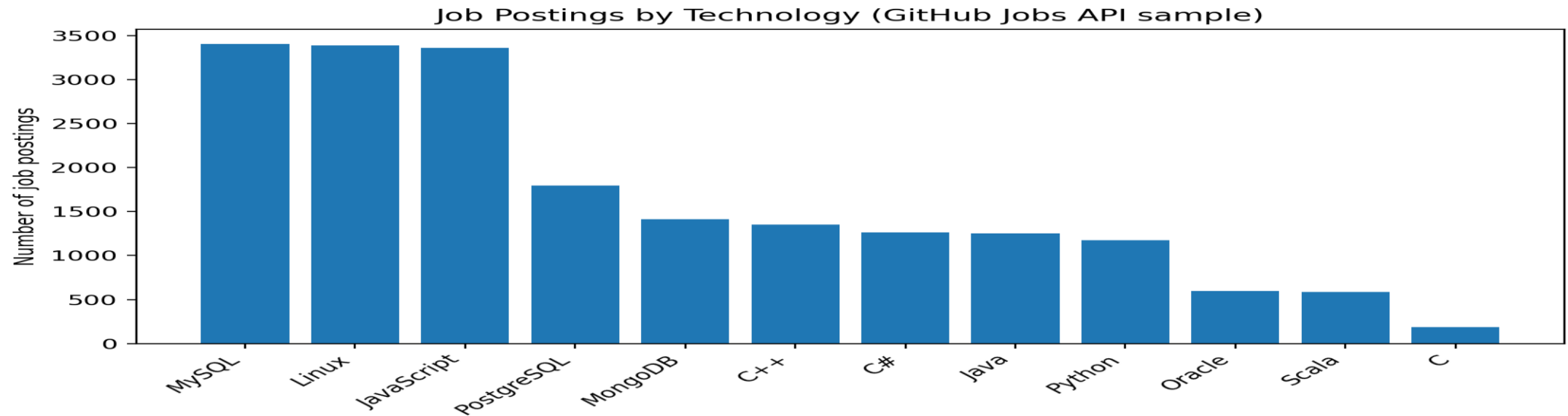
-  • Learners: SQL on PostgreSQL + one cloud warehouse exposure (BigQuery/Snowflake).
-  • Teams: plan governance for diverse DB usage (security, cost, data quality).
-  • Hiring: test fundamentals (SQL, modeling) over tool checklists.

Impact

- Highest-ROI skill: SQL + data modeling transfers across every database.
- Analysts benefit from Postgres fluency plus awareness of Redis-style caching in product contexts.



JOB POSTINGS



Market validation (job-postings.xlsx)

- Demand concentrates around MySQL/PostgreSQL + Linux + JavaScript (full-stack + infrastructure).
- Python appears in the requested set, but lower than web/infra signals in this sample.
- **Positioning:** highlight SQL + Linux fundamentals + one language aligned to your target role.
- **Portfolio:** build a SQL-driven dashboard/ETL project that proves analysis + delivery end-to-end.



DASHBOARD



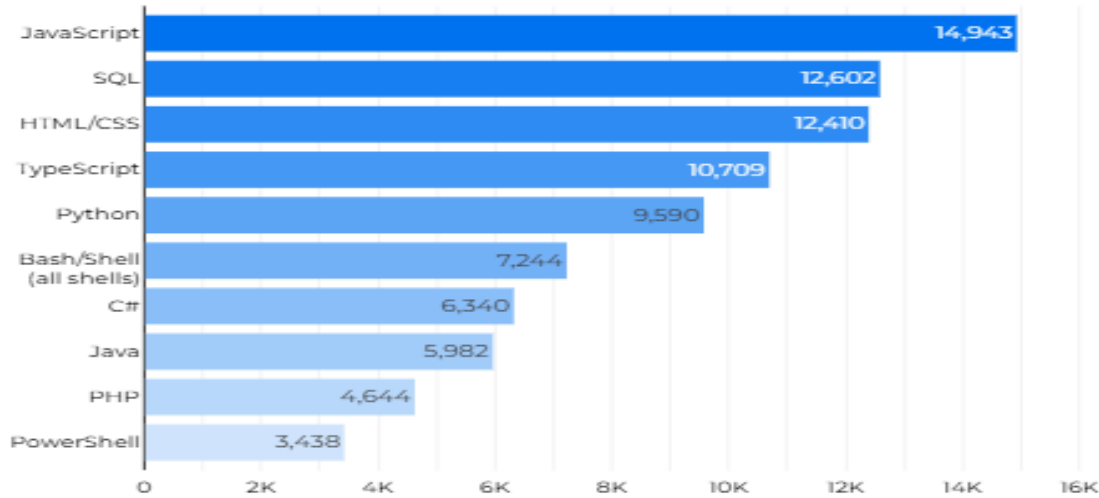
Dashboards were created in Looker Studio to make findings explorable and easy to validate.

- • Tab 1 — Current Technology Usage: languages, databases, platforms, frameworks used today.
- • Tab 2 — Future Technology Trends: next-year demand signals for the same categories.
- • Tab 3 — Demographics: who responded (age, country, education) to interpret results responsibly.
- Slides 12–14 show full screenshots of each tab.

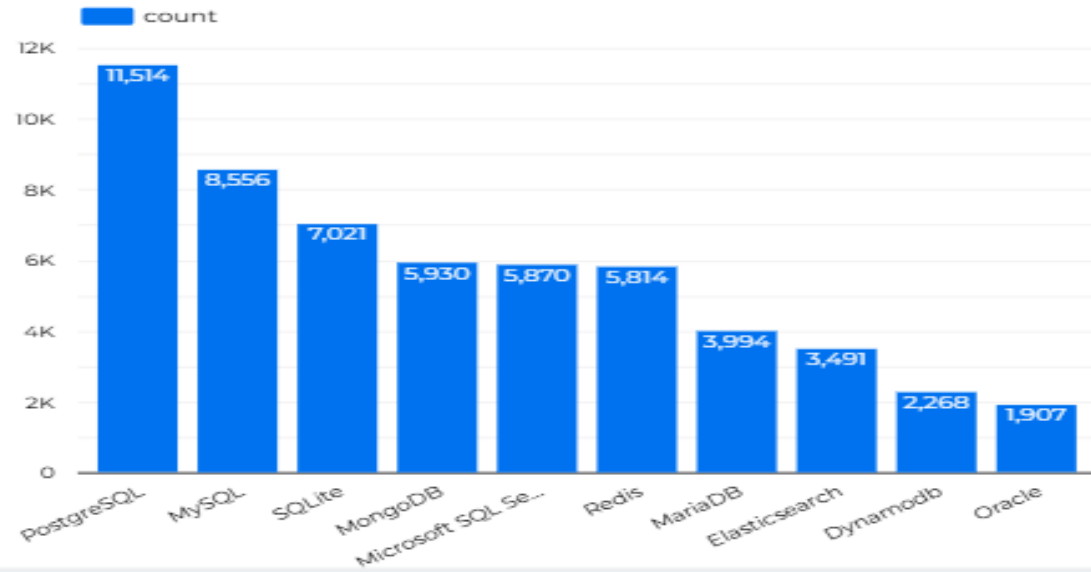
DASHBOARD TAB 1

Current Technology Usage

Top 10 Languages Worked With



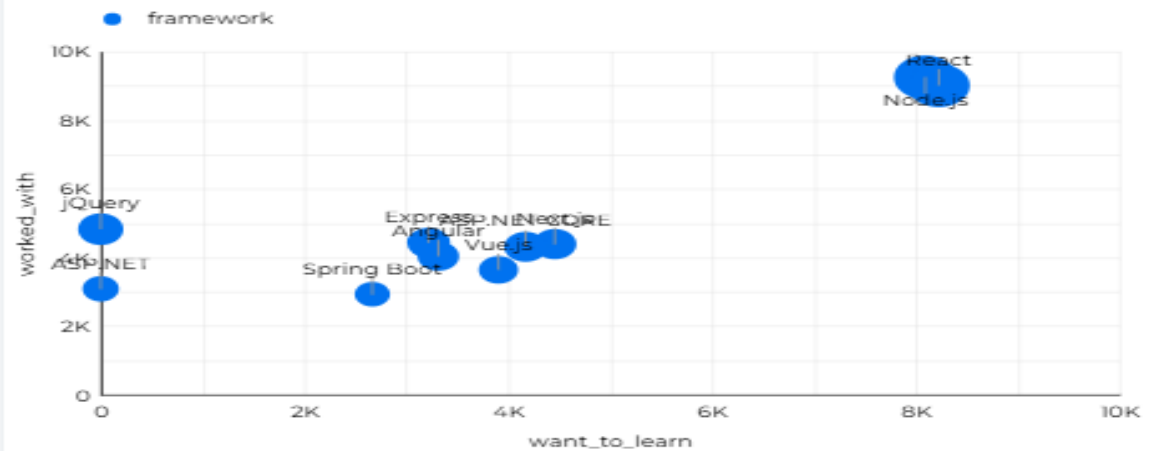
Top Databases Worked With



Top 10 Platforms

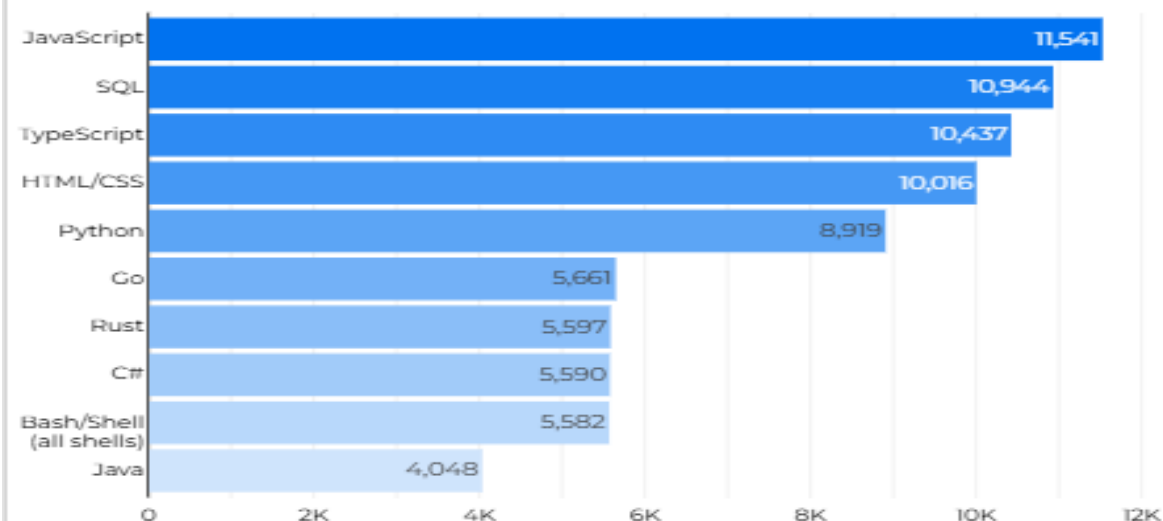


Top 10 Frameworks Used and Want to Learn With

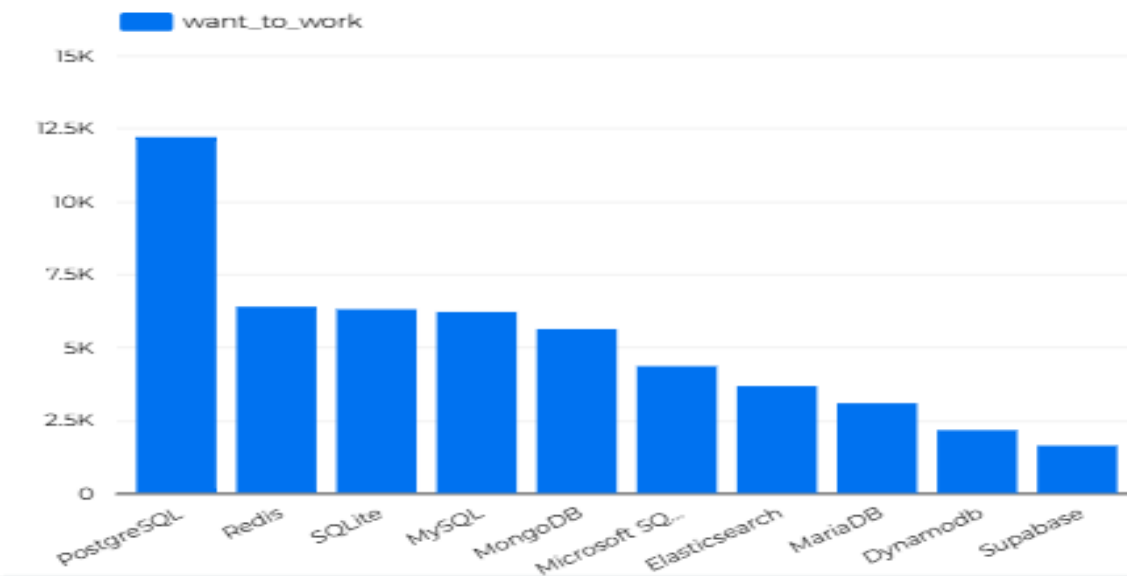


DASHBOARD TAB 2 – Future Technology Use

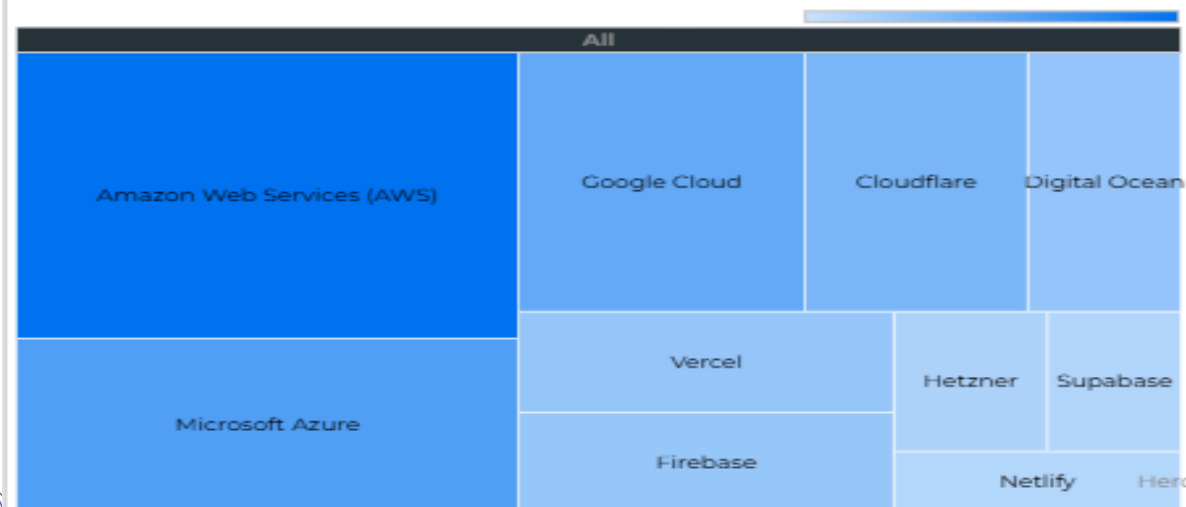
Top 10 Languages Desired Next Year



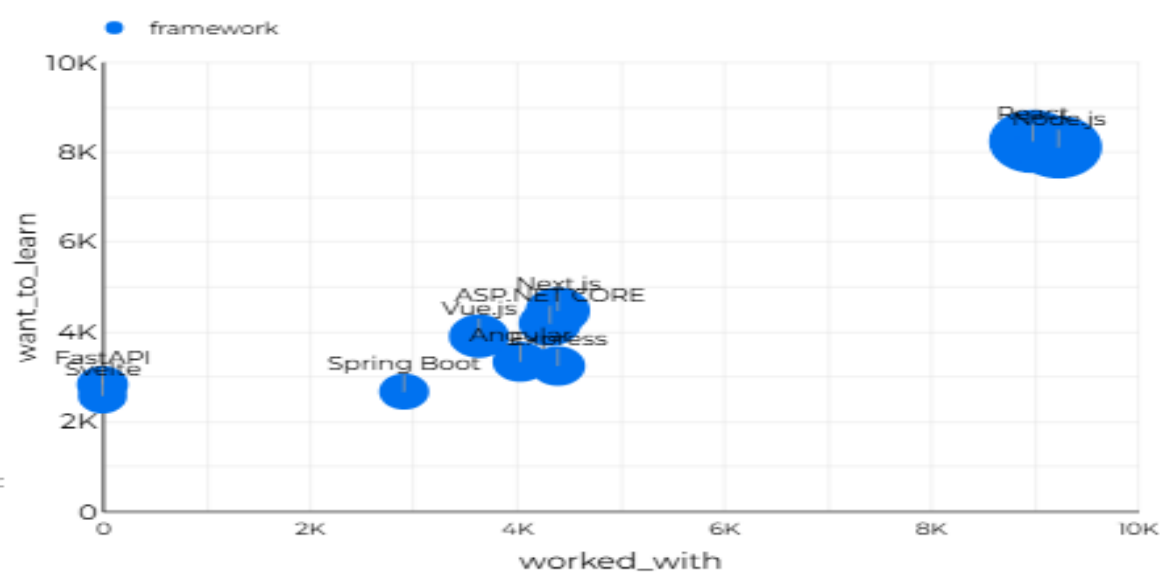
Top 10 Databases Desired Next Year



Top 10 Platforms Desired

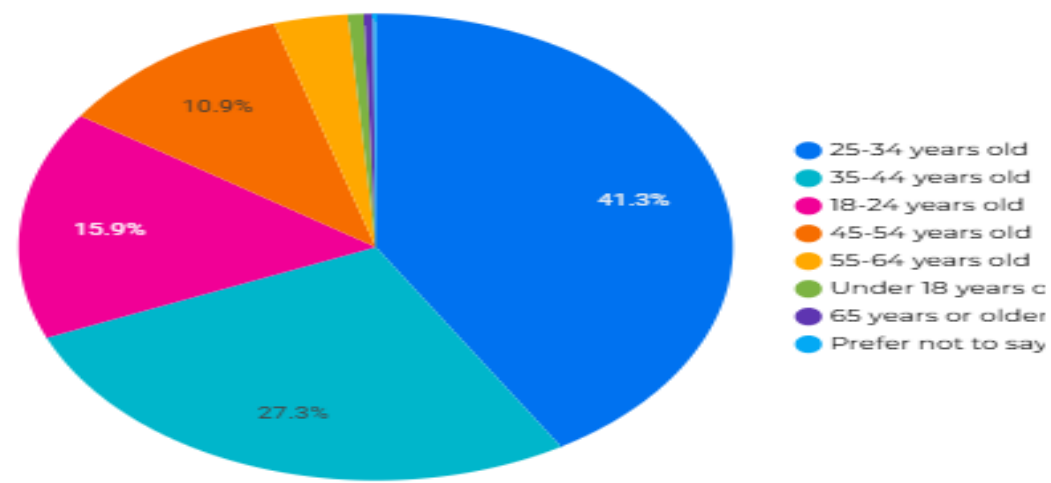


Top 10 Desired Web Frameworks: Worked With vs Want to Learn

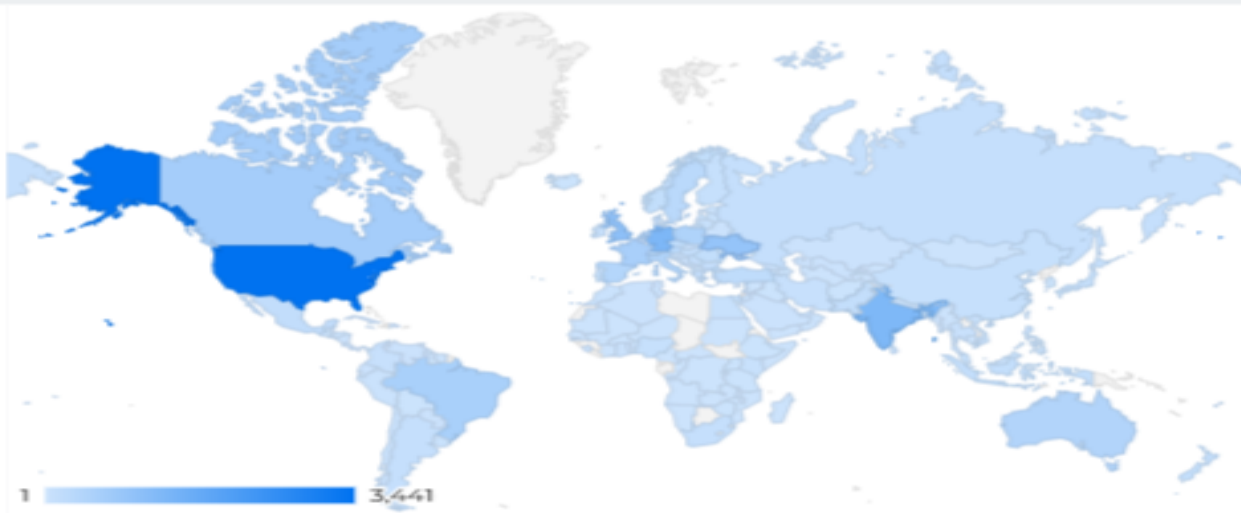


DASHBOARD TAB 3

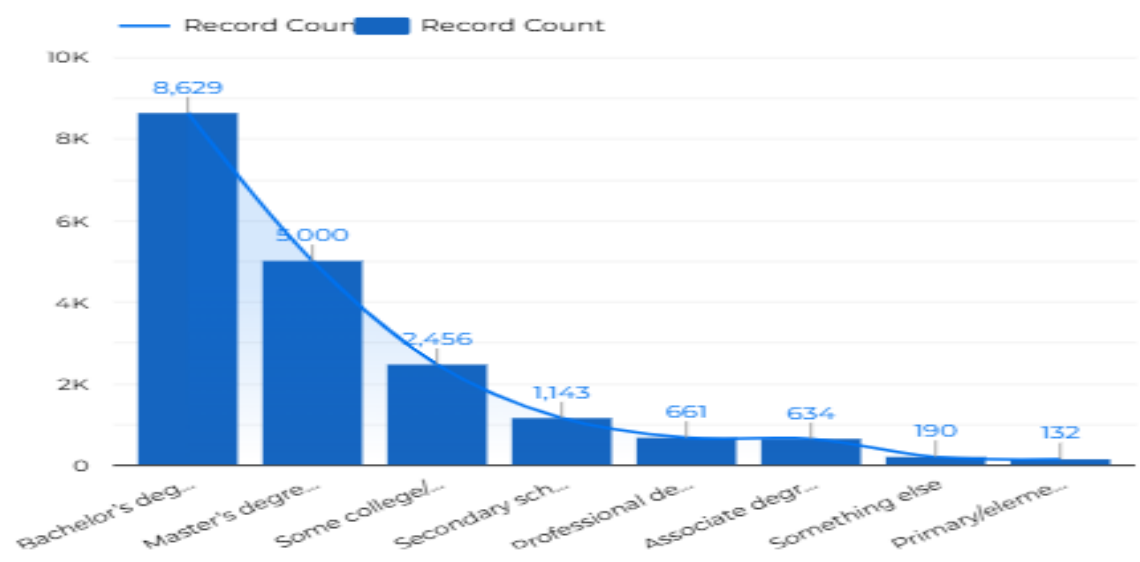
Age by Respondent Count



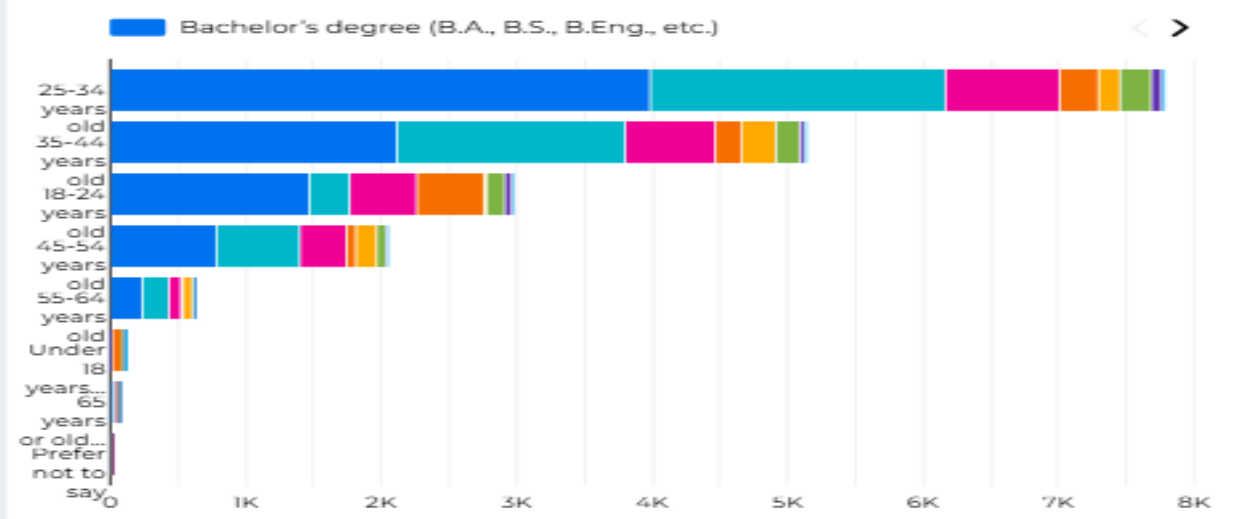
Respondent Count by Country



Respondent Distribution by Education Level



Respondent Count by Age and EdLevel



INSIGHTS FROM DASHBOARD

- Core stack dominates today + tomorrow: JS/TS + SQL + Python remain Top-5 in both current and desired rankings.
- Modernization shows up in “desire” metrics: TypeScript > usage; Go/Rust interest increases; Redis rises → more scalable, real-time systems.
- Cloud is baseline: AWS leads both current + desired platforms; Azure/GCP remain strong → cloud literacy is table-stakes.
- Demographics support production relevance: responses skew mid-career (25–44) and formal education → trends reflect workplace usage.
- **Action roadmap:** Build SQL + Postgres + Python + BI, then specialize (Product: JS/React; Data: modeling; Backend: Go/Rust).



OVERALL FINDINGS & IMPLICATIONS

Key findings

-  • Durable default stack: SQL + PostgreSQL + JavaScript/TypeScript, with Python as an analytics multiplier.
-  • Demand signals show modernization (TypeScript, Redis) and growth bets (Go/Rust) beyond the core.
-  • Job-postings snapshot aligns with survey: SQL databases + JavaScript + Linux are consistently requested.

• Implications

- **Individuals:** master the core first, then add one specialization aligned to your target role.
- **Teams:** upskill toward typed + cloud-native + real-time data patterns.
- **Education:** prioritize end-to-end, portfolio-ready project work.

Recommended next steps

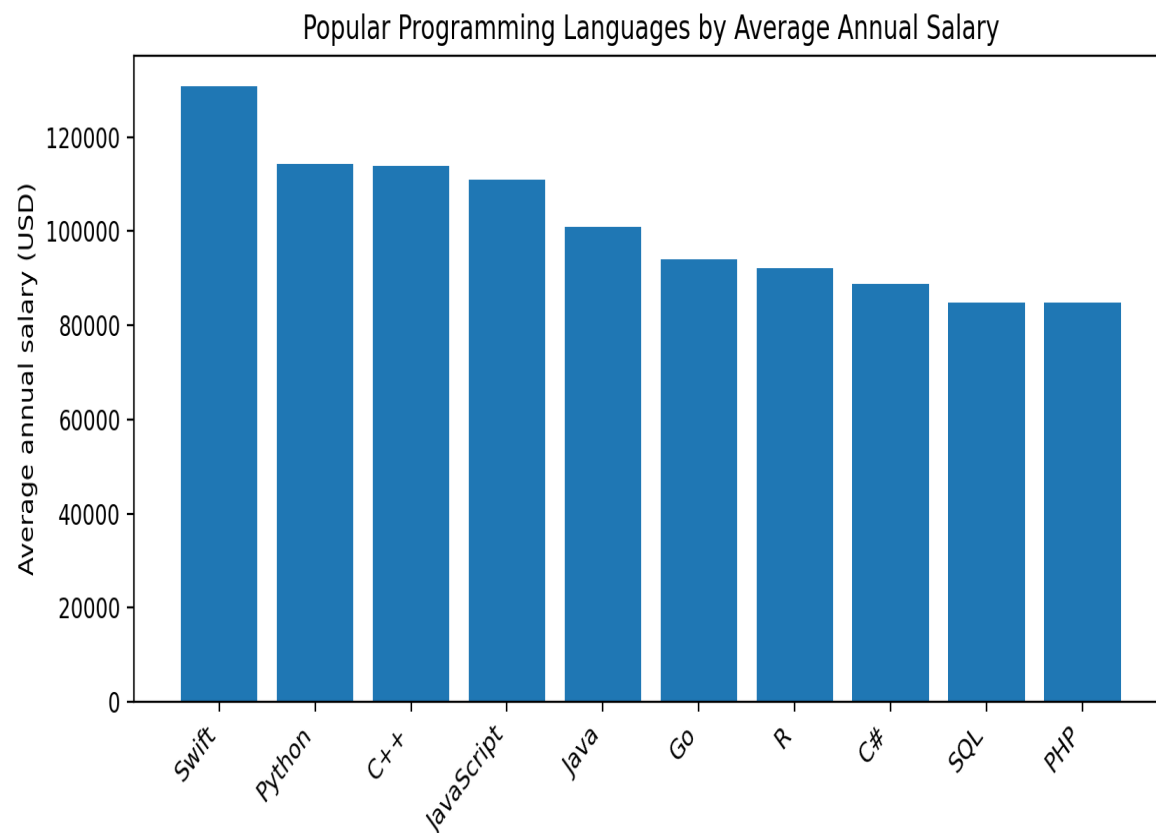
-  • Build 2 artifacts: (1) SQL+Python notebook analysis, (2) interactive dashboard with clear KPI story.
-  • Add cloud baseline: load one dataset into a cloud warehouse and document permissions/costs at a high level.
-  • Write a 1-page decision memo: “What we found → so what → what to do next.”

CONCLUSION



- 📌 The market has a durable core stack: JavaScript/TypeScript + SQL + Python, backed by PostgreSQL/MySQL.
- 📈 Future demand stays aligned: teams modernize toward typed JavaScript and cloud-first data workflows.
- 🧠 Dashboards confirm a “winner-takes-most” pattern and allow segmented analysis by age/country/education.
- ✅ Action plan: master SQL + Python + BI; then specialize (cloud data, analytics engineering, or web analytics).

APPENDIX — Additional Analysis & Visuals



- **Appendix highlights**
- **Job postings by technology (job-postings.xlsx) — market validation signal.**
- **Popular languages vs salary — triangulates demand with compensation.**
- *Limitations*
- **Survey is self-reported; multi-select counts represent mentions; results are directional indicators.**

 Additional supporting visuals from the notebook (non-required).

 Appendix A — EDA Overview (Notebook checks)

What was cleaned + why these checks matter

 Appendix B — Geographic Distribution & Remote Mix

Top countries and how work arrangement varies

 Appendix C — Satisfaction Rating Distribution

Ratings cluster (7–8/10) → opportunity to improve tooling/UX

Appendix D — Compensation Distribution (Trimmed)

Pay is right-skewed; median is more representative than mean

Appendix E — Experience vs Compensation (Trimmed)

Experience explains some pay but variance stays high

Appendix F — AI Usage: Current vs Interested

Where AI is already used vs biggest growth runway

Appendix G — AI Adoption Gap Index

Prioritize skills with highest Want–Have gap

Appendix H — Education & Age Segmentation

Checks that trends persist across cohorts

Appendix I — Technology Usage Snapshot

Confirms core stack: JS/TS + SQL + Python + Postgres; AWS/Azure lead



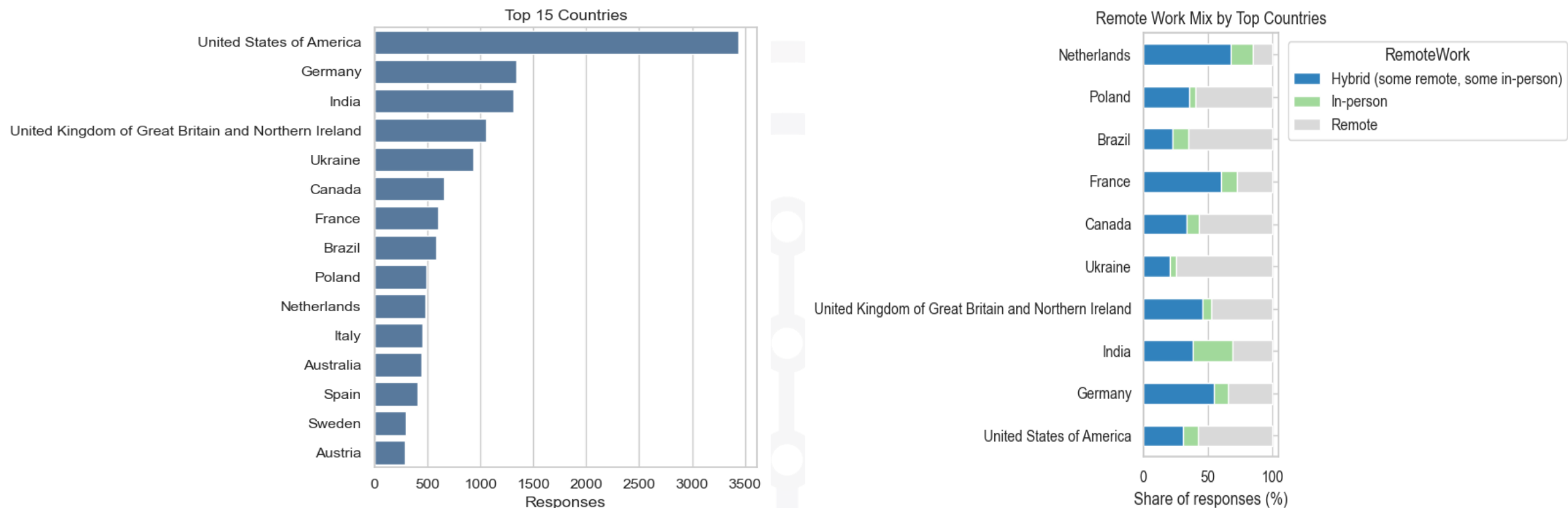
APPENDIX A — Notebook Exploration (EDA Overview)

 Additional EDA visuals to support deeper insight beyond the required slides.

 Goal: validate patterns seen in dashboards and quantify real-world implications (pay, remote work, and AI tooling).

 All charts are derived from the same Stack Overflow survey dataset used throughout the capstone.

APPENDIX B – Geographic Distribution & Remote Work Mix



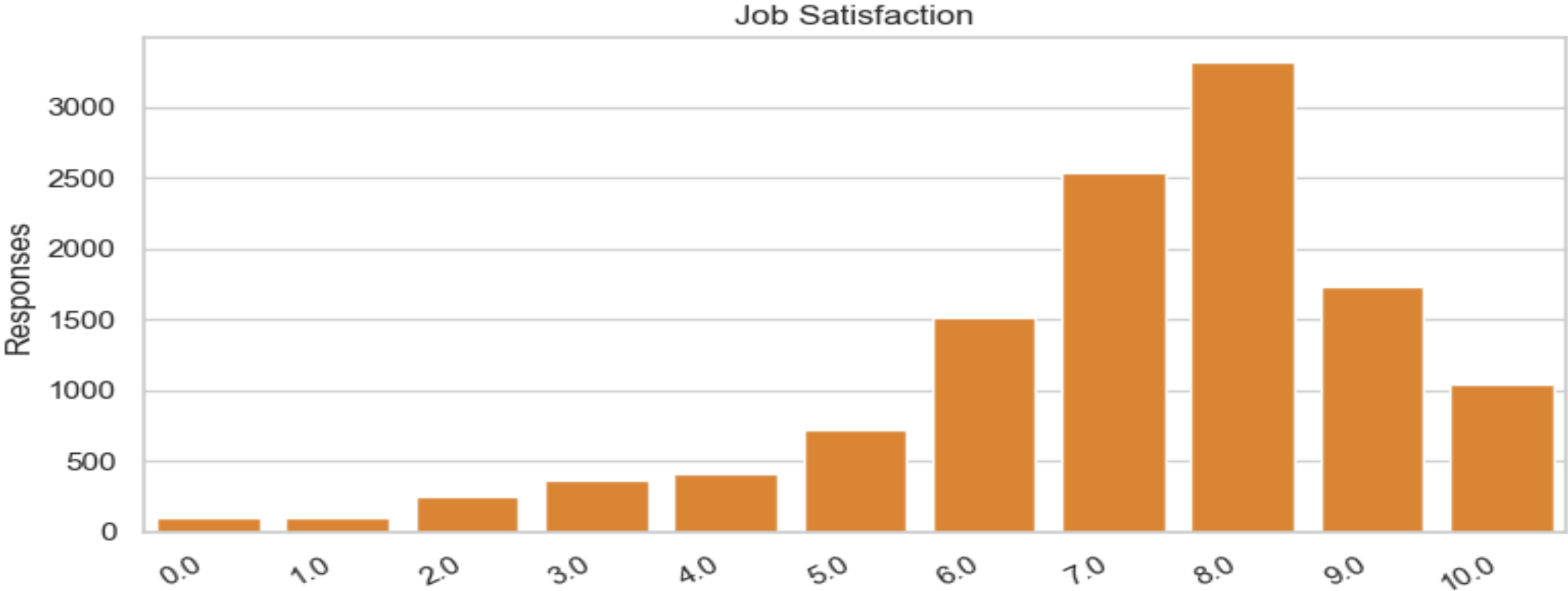
 Top 15 respondent countries

 Remote work mix within top countries

 The survey sample is global but skewed toward a handful of countries (US and India lead), so global “top 10 tech” signals are influenced by these major markets.

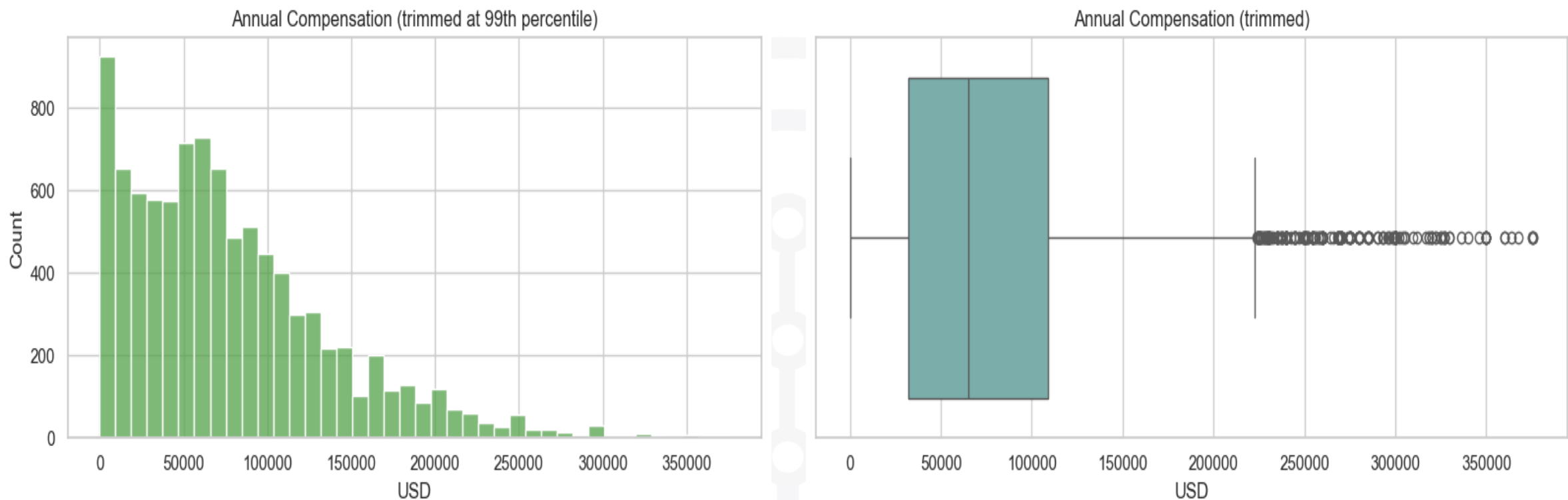
 Remote work is common across top countries, but the mix differs—some regions show higher hybrid/on-site shares, which affects compensation expectations and hiring strategy.

APPENDIX C – Job Satisfaction Snapshot



-  Ratings cluster around 7–8/10, suggesting many developers feel reasonably positive about their current role (but there is still a meaningful “at-risk” tail below 6).
-  For employers: improving flexibility, growth paths, and tooling quality likely delivers outsized retention ROI compared to compensation-only adjustments.

APPENDIX D – Compensation Distribution & Outliers



 Histogram: annual compensation (trimmed)

 Box plot: annual compensation (trimmed)

 Compensation is heavily right-skewed: most respondents fall in the lower-to-mid range, while a smaller number of high earners create a long tail.

 Implication for analysis: medians and percentiles are better than averages for “typical pay,” and segmentation (role, experience, country) matters more than global means.

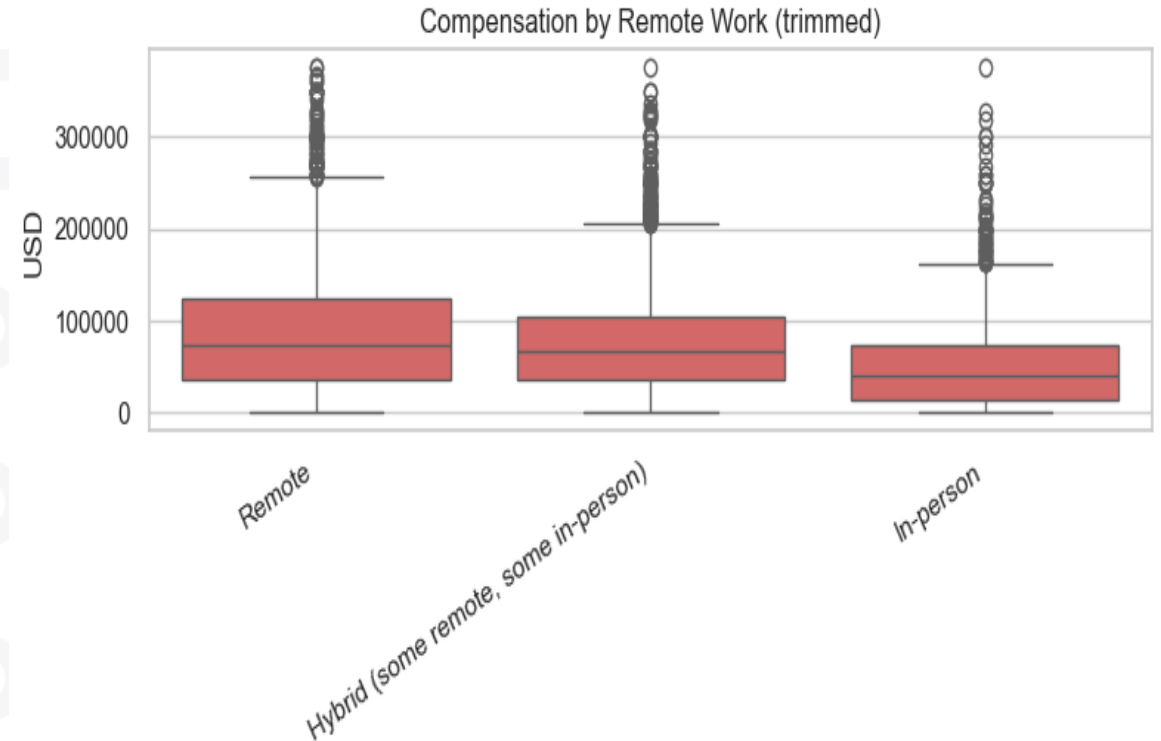
📌 APPENDIX E — Drivers of Compensation: Experience & Work Arrangement



📌 Experience vs annual compensation (trimmed)

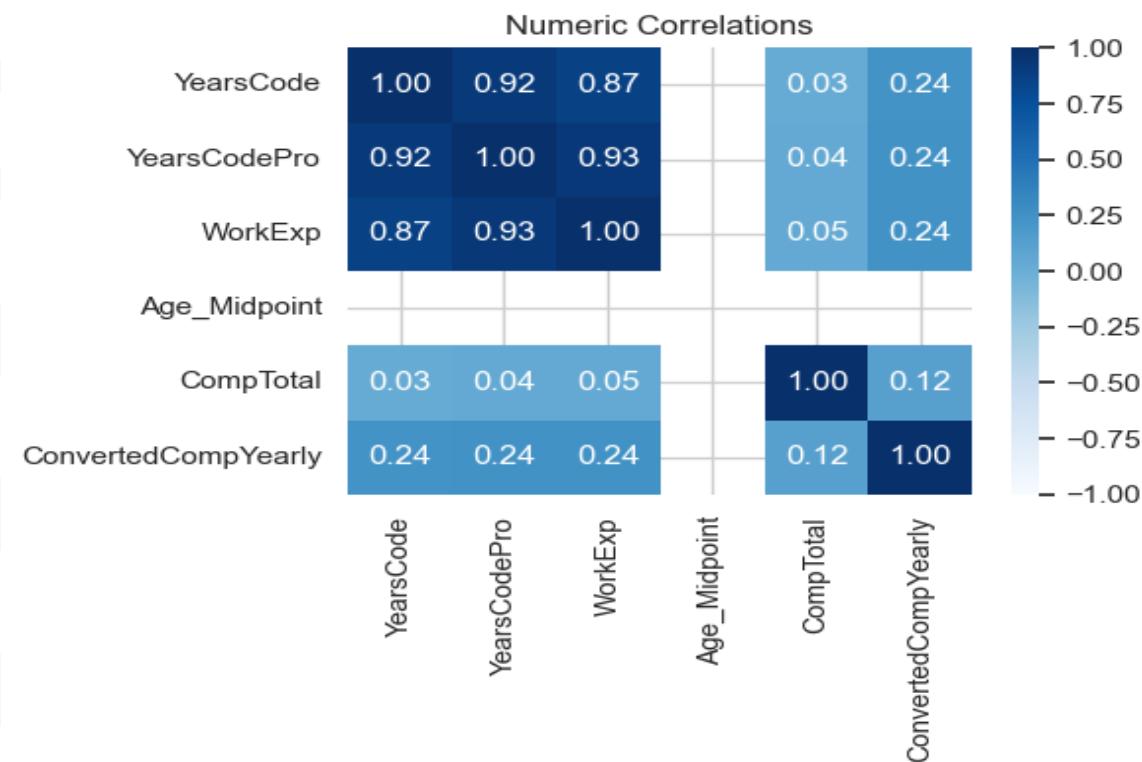
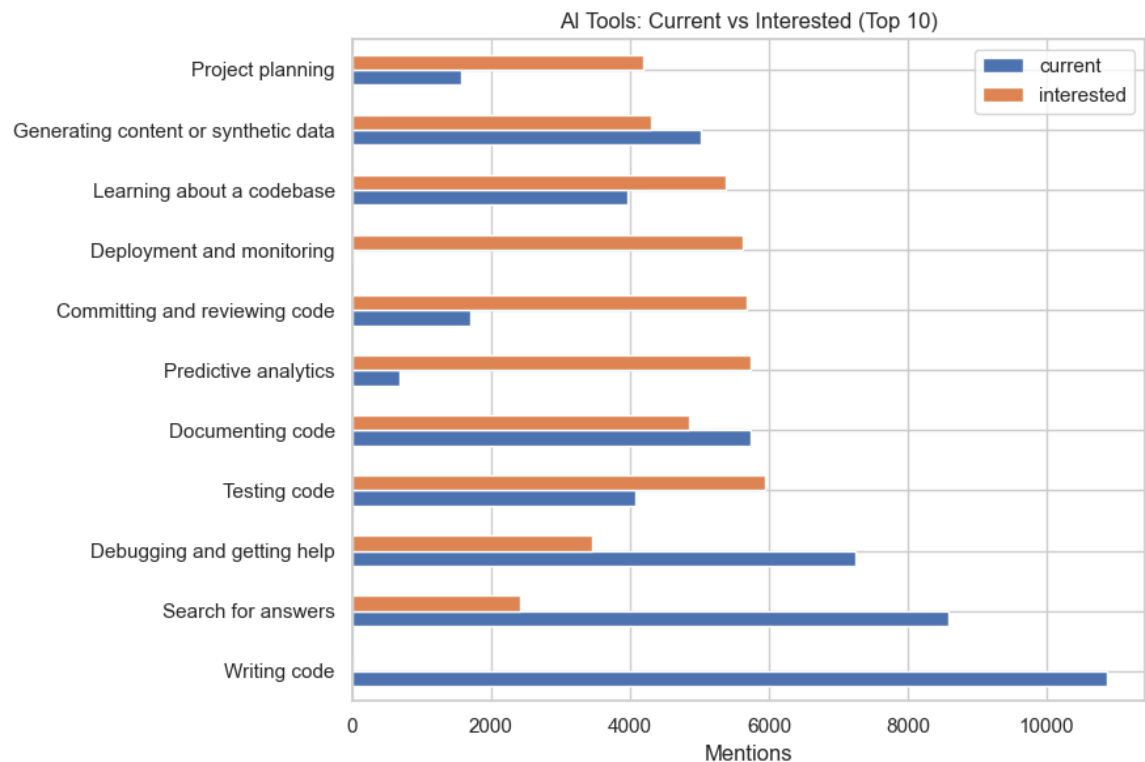
📌 Years of experience shows a clear upward relationship with compensation, but with wide dispersion—role, industry, and geography likely explain the spread.

📊 Remote/hybrid work is associated with higher median pay in this sample; however, this may reflect confounding factors (senior roles skew remote).



📌 Compensation by remote work status (trimmed)

📌 APPENDIX F — AI Tool Adoption Gap & Skill Correlations



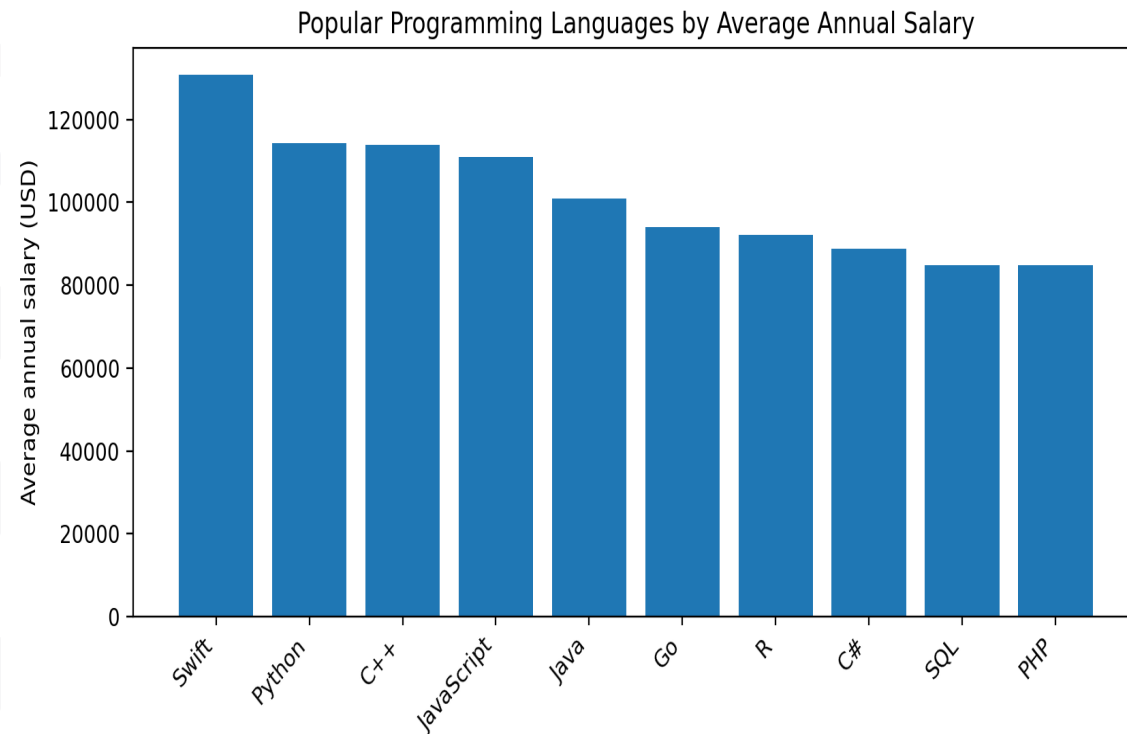
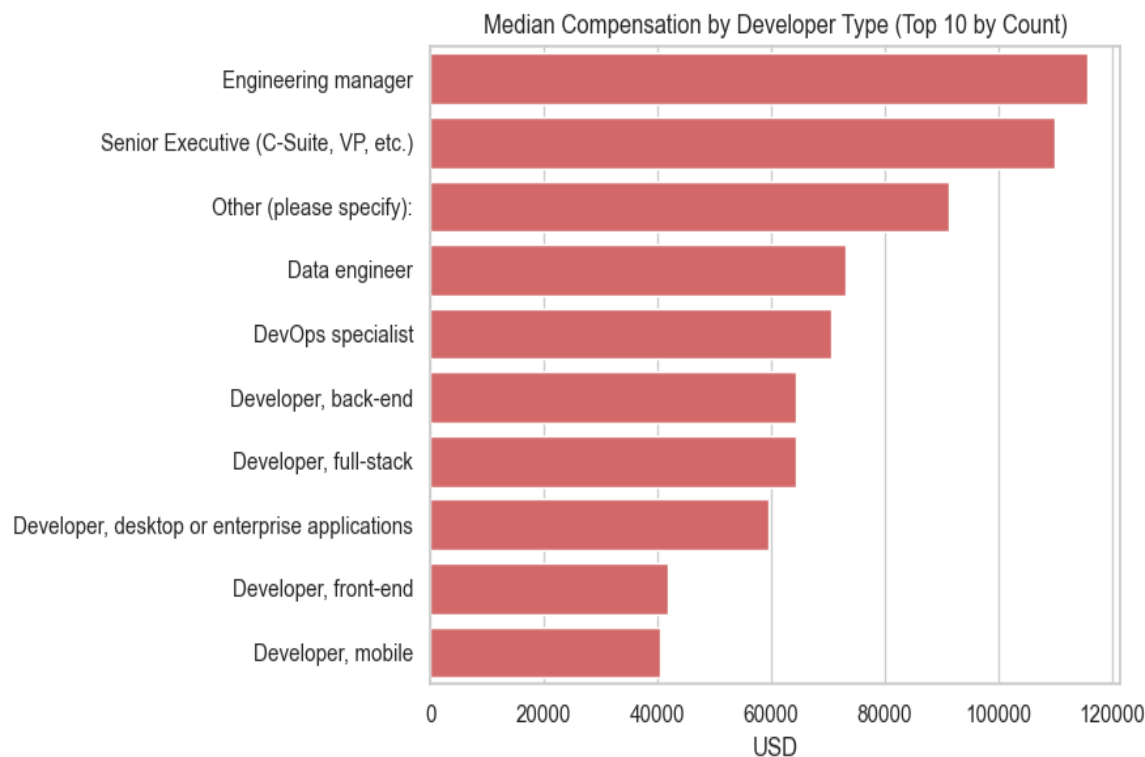
📌 AI tools: current use vs interest

📌 Numeric feature correlation heatmap

📌 Interest in AI tools outpaces current use—an adoption gap that suggests near-term demand for AI-augmented workflows (coding assistants, analytics copilots).

📊 Correlation checks help validate that key numeric fields behave as expected and support more advanced modeling/segmentation in future work.

APPENDIX G – Median Compensation by Role & Popular Languages



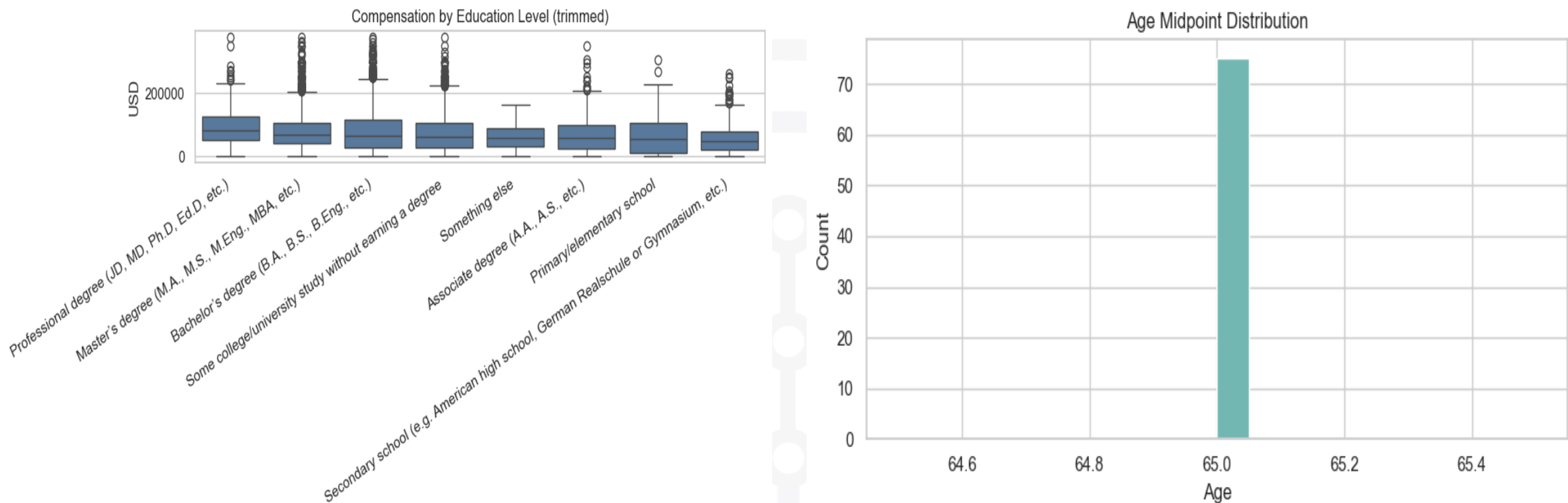
 Median pay by developer type (trimmed)

 Median pay by popular languages (trimmed)

 Leadership/management and specialized engineering roles tend to show higher median compensation, reinforcing that seniority + specialization are major pay levers.

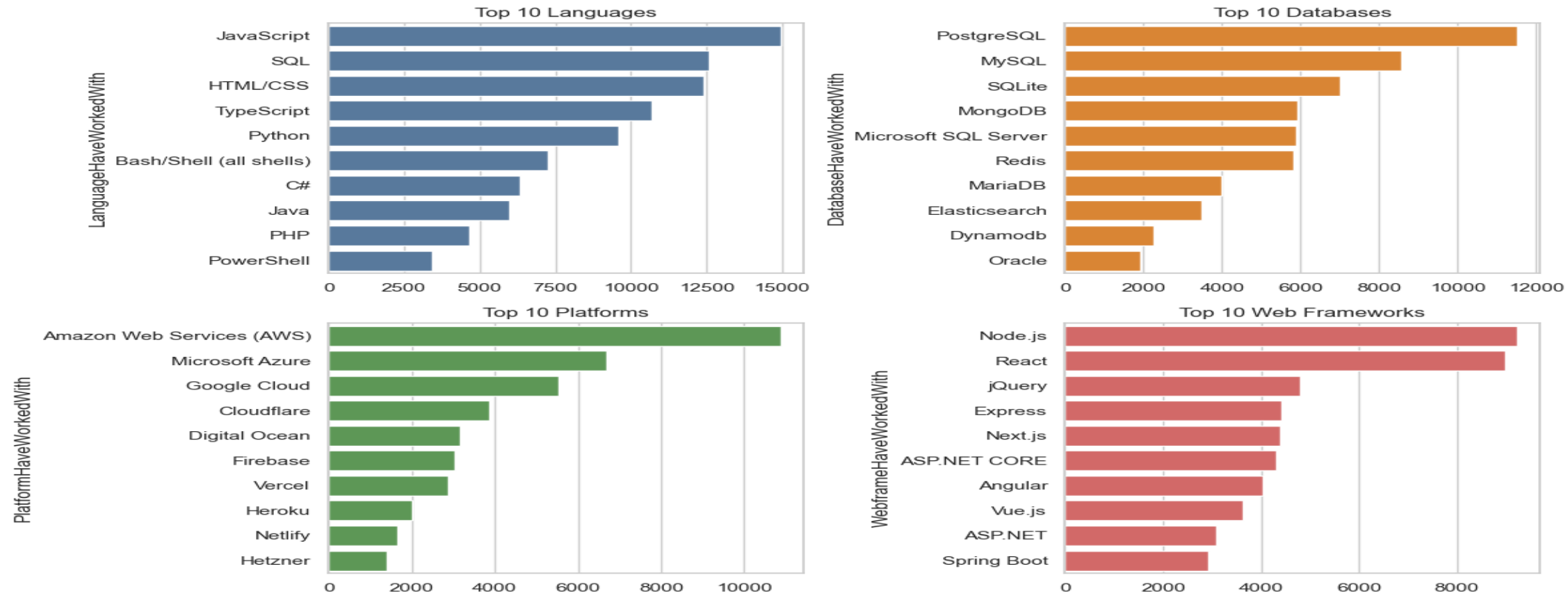
 Language signals correlate with market value: widely adopted ecosystems (SQL, Python, JavaScript) remain core, while niche/specialist stacks can carry premiums depending on demand.

📌 APPENDIX H — Education & Age: Segmentation Checks



- 📌 Compensation by education level (trimmed)
- 📌 Education shows a visible uplift in median compensation at higher attainment levels, but the spread is large—experience and role can outweigh degrees.
- 📌 Age midpoint chart highlights why age-band analysis (used in dashboards) is often more robust than converting ranges to a single numeric midpoint.
- 📌 Age midpoint distribution (imputation check)

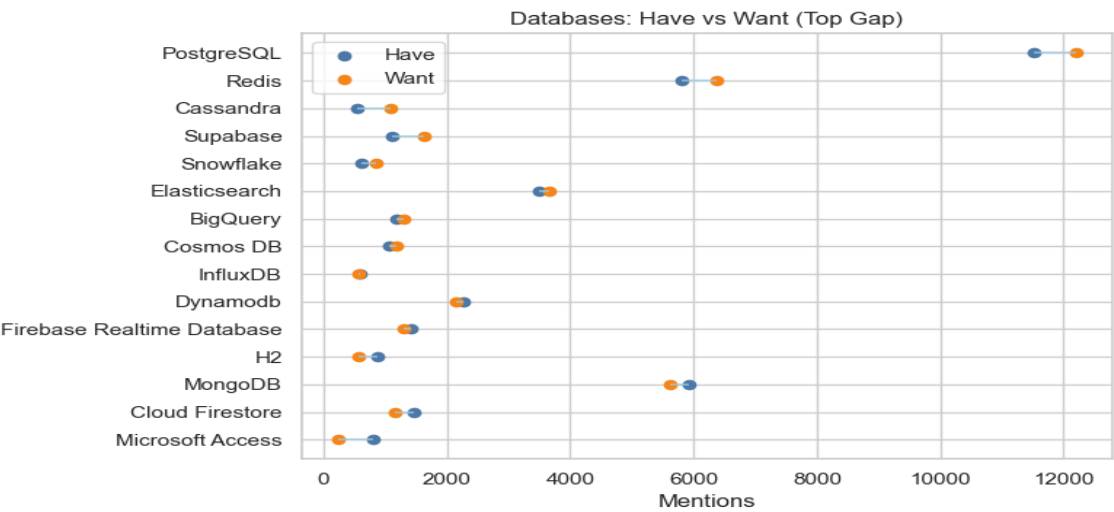
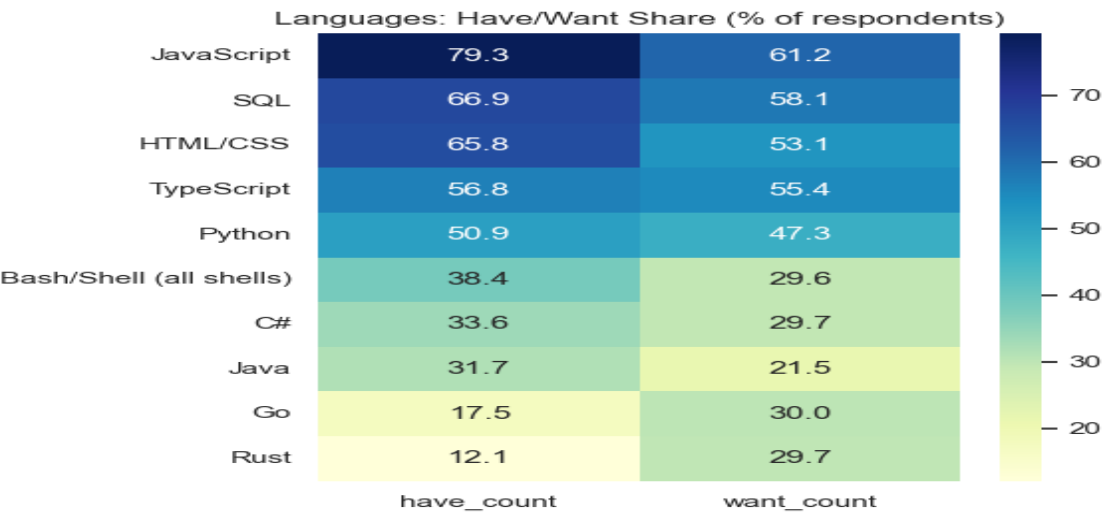
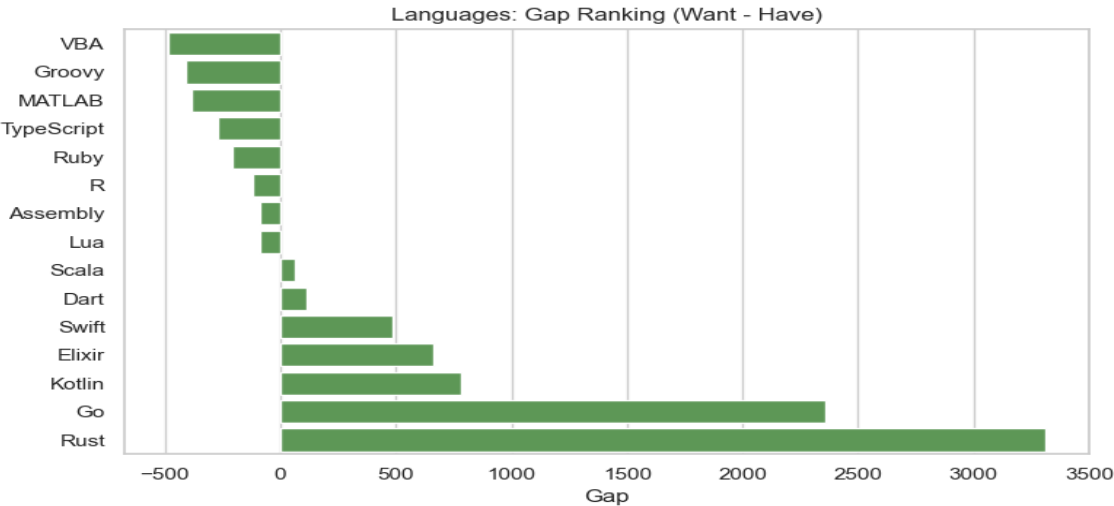
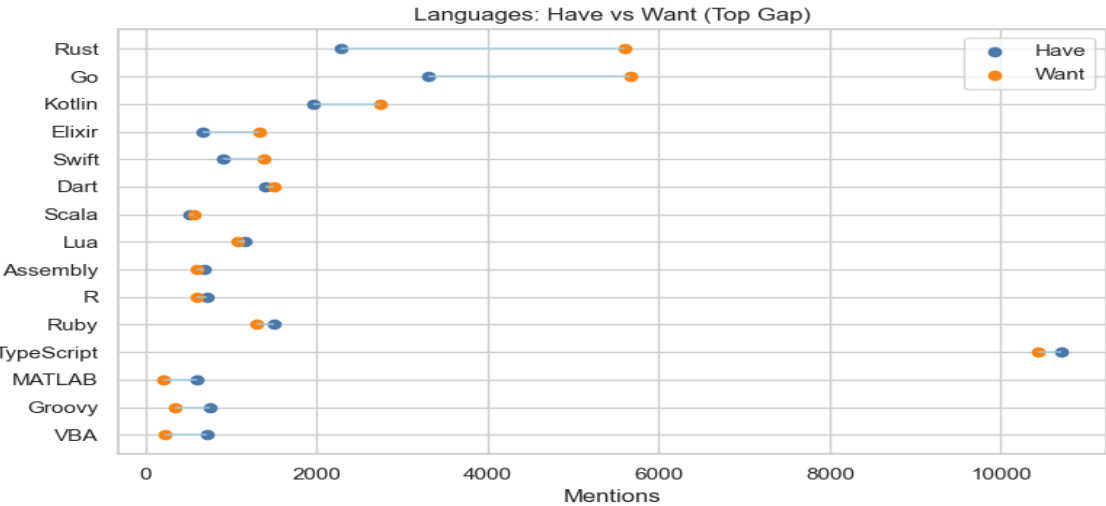
📌 APPENDIX I – Technology Usage Snapshot (Notebook EDA)



📌 This consolidated view matches the main Results slides: JavaScript/SQL/Python dominate, PostgreSQL/MySQL lead databases, and AWS/Azure lead platforms.

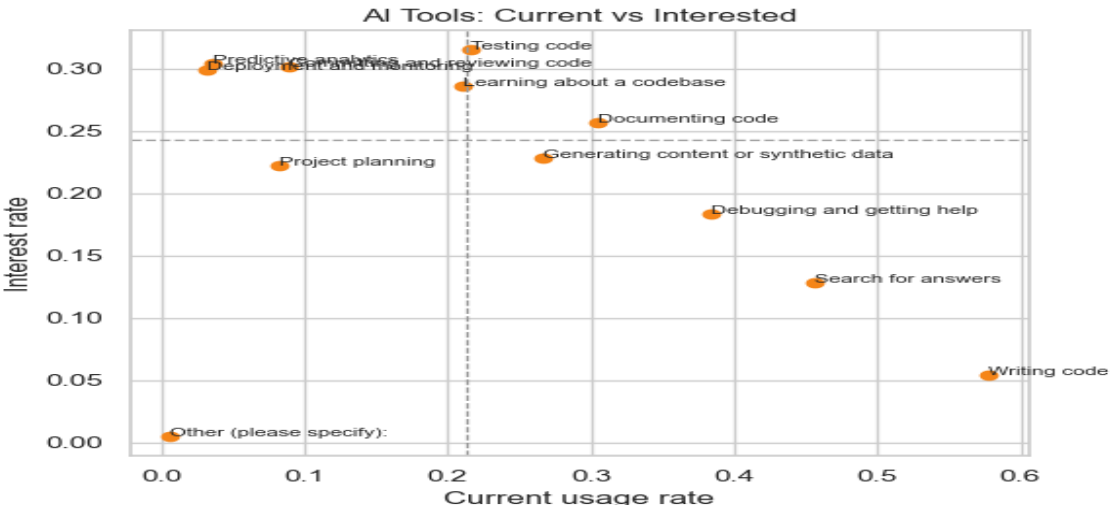
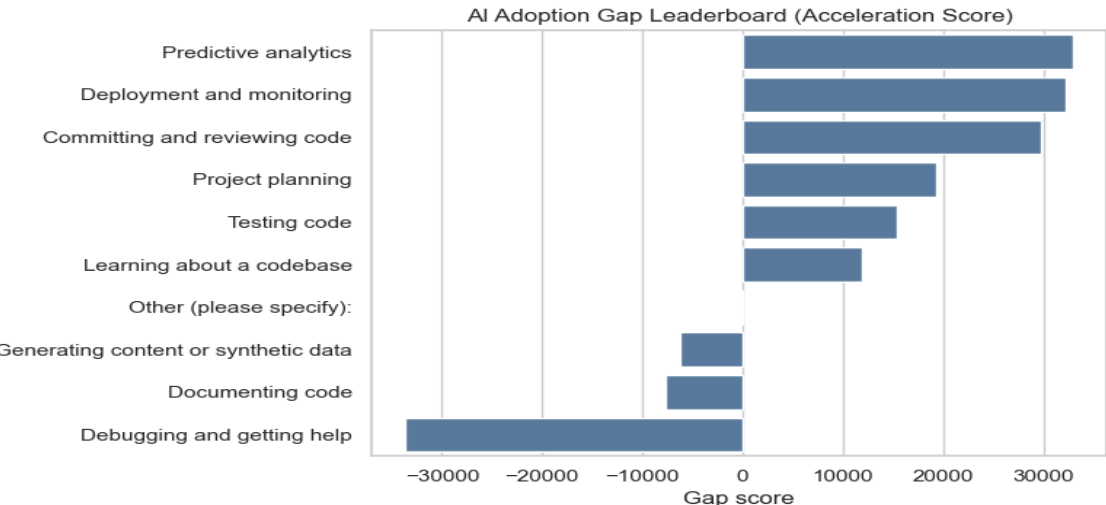
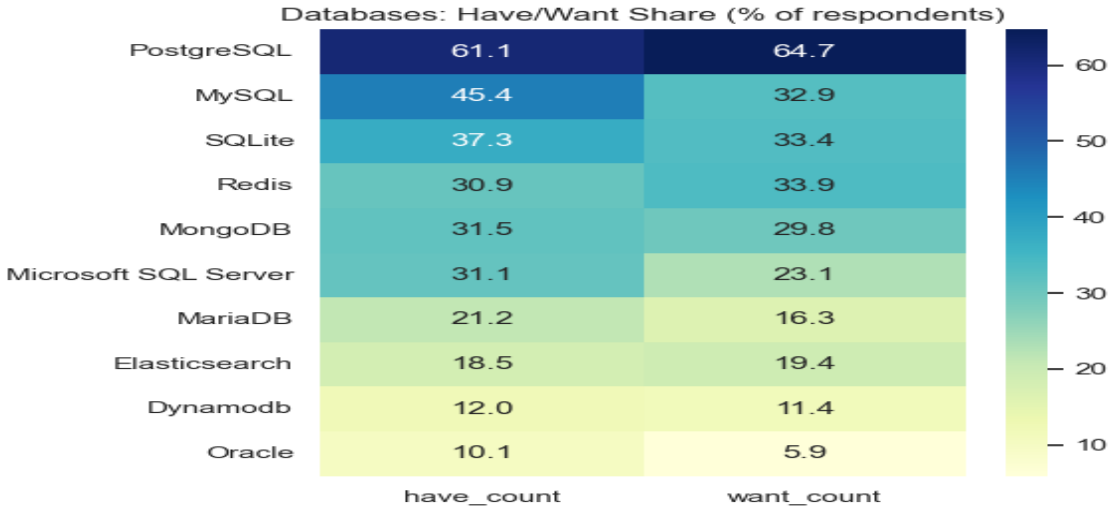
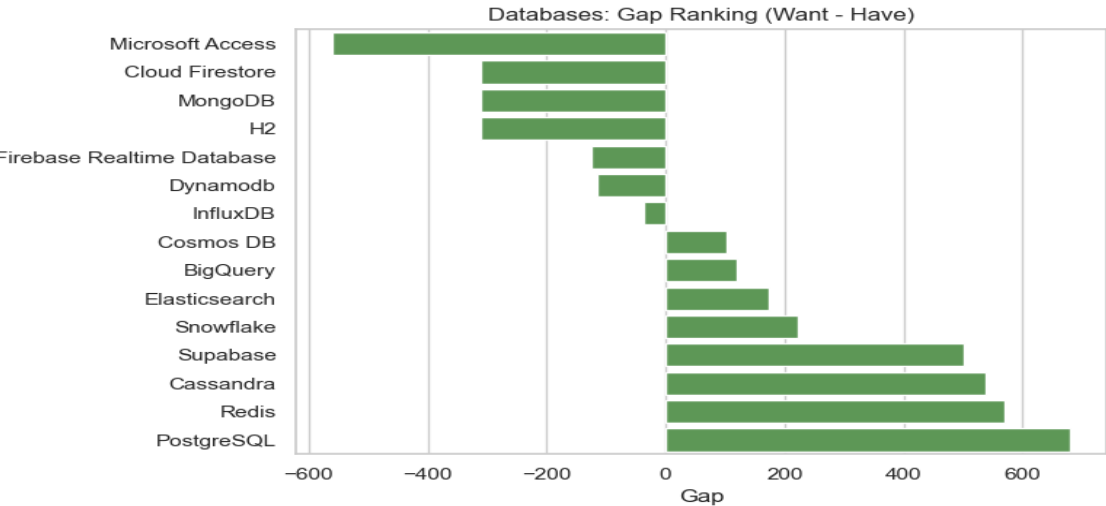
📊 Framework demand clusters around React/Node/Next.js—useful guidance for learners prioritizing job-relevant web stacks.

APPENDIX J — Additional Supporting Visuals (Notebook)



Note: These segmented views help validate that the core trends hold across slices; use for deeper discussion if asked.

APPENDIX K – Additional Supporting Visuals (Notebook)



Note: Included as optional evidence; not required for grading, but strengthens the depth of analysis.

Recommendations & Action Plan

 What to do with these findings (skill strategy + hiring strategy).

For Learners / Career Pivot

 Prioritize durable core stack: SQL + Python + BI + one cloud (AWS/Azure).

 Add a web “translator” layer: JavaScript basics + REST APIs to collaborate with product/engineering.

 Invest in React/Node/Next.js only if you want web dev roles—otherwise focus on analytics depth.

 Build portfolio proof: 2–3 end-to-end notebooks + dashboards + short exec memos (this capstone format). Use AI tools for speed, but validate outputs—treat AI as a productivity multiplier, not a substitute.

 Bottom line: The market rewards broad adoption (SQL/Python/JS) + cloud fluency, while specialization and seniority drive the highest compensation.

For Employers / Hiring Teams

 Expect SQL as baseline; Python is a major differentiator for automation + advanced analysis.

 Cloud literacy is becoming standard even outside DevOps—AWS/Azure exposure improves team agility.

 React/Node/Next.js demand signals remain strong; hire for ecosystem fluency, not just one framework.

 Remote work remains a talent lever; align comp bands with role seniority + location realities. Support AI-augmented workflows (policy + tooling) to boost throughput without sacrificing quality.

Next Research Questions (Optional Extensions)

 How to extend this work into a senior-grade analysis (segmentation + modeling).

 Compensation by role × country: normalize by cost-of-living and compare medians/percentiles (not averages).

 AI adoption index: quantify “use vs interest” gap per role/experience to identify fastest-moving segments.

 Job satisfaction drivers: model satisfaction as a function of remote status, role, AI usage, and compensation (log-transform).

 Tech interest gap analysis: compute $\Delta = \text{want} - \text{have}$ for languages/databases/frameworks to highlight upskilling demand.

Cohort analysis: compare early-career (18–24) vs mid-career (25–44) vs senior to see diverging stack priorities.

Hiring implications: identify the smallest skill set that covers the widest job market (skills coverage optimization).

Data quality follow-up: evaluate missingness patterns and sampling bias to strengthen the credibility of conclusions.



Key Takeaways (1-Slide Recap)

📌 What the market is using now, what it wants next, and how to act on it.

📌 Core stack is stable: JavaScript + SQL + Python remain the highest-coverage skills across roles.

📊 Data backbone winners: PostgreSQL leads both current use and next-year interest → strong “safe bet” database.

🧠 Cloud + web ecosystems dominate: AWS/Azure lead platforms; React + Node/Next.js cluster as top frameworks.

✅ AI tool interest exceeds current adoption → near-term advantage for teams that operationalize AI-assisted workflows.

📌 Action: For job-seekers → build portfolio projects in SQL/Python + dashboards. For teams → standardize on cloud + AI tooling for faster delivery.

